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BSIT 2-3

Activity #1 - Variables

Instructions:

- Carefully analyze the given Java program.
- Trace the references and memory changes step-by-step.
- Complete all required tables and questions.
- Show your logical reasoning where necessary.
- Write your answers in a yellow paper (handwritten).
- Take a clear photo or scan of your work.
- Upload your file in Canvas on or before the deadline.
- Submission Format: BSIT 2-3_Caparas_Ederlyn_codeAnalysisActivity#1.pdf

```
class Account {
    String owner;
    double balance;

    Account (String owner, double balance) {
        this.owner = owner;
        this.balance = balance;
    }
}

public class TestAccount {
    public static void main(String[] args) {
        Account acc1 = new Account("Carlos", 5000);
        Account acc2 = new Account("Diana", 7000);
        Account acc3 = acc1;
        acc2.balance = 8000;
        acc3.owner = "Miguel";

        System.out.println(acc1.owner + " " + acc2.balance);
        System.out.println(acc2.owner + " " + acc2.balance);
        System.out.println(acc3.owner + " " + acc3.balance);
```

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Please answer the following questions:

PART A - VARIABLE IDENTIFICATION TABLE

Variable Name	Data Type	Variable Type	Stored In
acc 1	Account	Reference / Local	Stack (Points to Heap Object)
acc 2	Account	Reference / Local	Stack (Points to Heap Object)
acc 3	Account	Reference / Local	Stack (Points to the line Heap object as acc1)
Owner	String	Instance field	Heap (Inside Account object)
balance	double	Instance field	Heap (Inside Account object)

PART B: Explanation. Please see example below.

Statement	What Happens?	Memory Impact
ID Account acc1 = new Account ("Carlos", 5000);	ID this creates a new account object with owner "carlos" and balance 5000, acc1 which a local variable that stores the reference.	ID this stored into the heap new object allocated. Again it's stack; reference is stored.
ID Account acc2 = New account ("Diana", 7000);	ID this is the same as acc1 since it creates another Account object n;th owner "Diana" n;th a balance of 7000. Acc2 stores the reference	Again this another Heap or another object which is allocated. Again it's stack; reference is stored.
Account acc3 = acc1;	This just basically points to acc1 and which just basically point at the same objects as acc1 and in turn just replicates	the memory is unchanged since there is no new object both local variable points to the same heap object.

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acc2.balance = 8000;	This just updates the balance of acc2	Heap it just indicates the objects balance.
acc3.Owner = "Miguel";	Now this changes the owner field of the object reference in acc3 and thus Carlos becomes Miguel.	

PART C - Output

Miguel 5000.0
Diana 8000.0
Miguel 5000.0

> It changed the owner of acc1 entirely.

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